a new cluster configuration. It might seem complicated for new users, but stay calm when trying to choose the right one. We highly recommend the *Best Practices for Cluster Configuration* linked in *Further reading* for guidance, especially if you are responsible for setting up compute for one or more teams. Let's talk about some of the compute options as they relate to ML and this book:

- **Multi-node versus single-node**: Multi-node is excellent for distributed projects (think Spark). Single-node is suitable for projects or workloads that are performed on the driver (think scikit-learn or pandas).
- **Access mode**: Some cluster configurations support Unity Catalog, and some don't. Choose a cluster that supports UC for the projects in this book.
- **Databricks Runtime (DBR) and Python packages**: The DBR is part of the managed service of Databricks. A DBR is a group of commonly used packages bundled and pinned to a release version and then versioned as a DBR. There are two significant variations of a DBR: standard and ML. Data engineering workloads often utilize the standard DBR. The ML DBR, as the name implies, is for DS and ML workloads. Think of the greatest hits of ML Packages – sklearn, pandas, numpy, and seaborn, as well as the latest CUDA libraries for GPU clusters, and so on, these are included in the ML runtime, which means they are pre-installed and ready to be used in your code. DBRs provide the benefit of reproducibility as well as reducing the engineering development time of packaging and pinning versions. When adding additional libraries, you can add them to your notebook, cluster, or workspace.

  The best practice for production is to install cluster-scoped libraries rather than notebook-scoped libraries. However, notebook-scoped libraries are great for interactive use and development (please note that if you opt for notebook-scoped libraries, every notebook will require you to re-install non-default libraries each time, as one notebook represents a unique `pyenv`). You will install libraries both at the cluster level and in a couple of project notebooks. You can see the **Libraries** tab in *Figure 2.13*. Simply click **Install new** to install a new library.

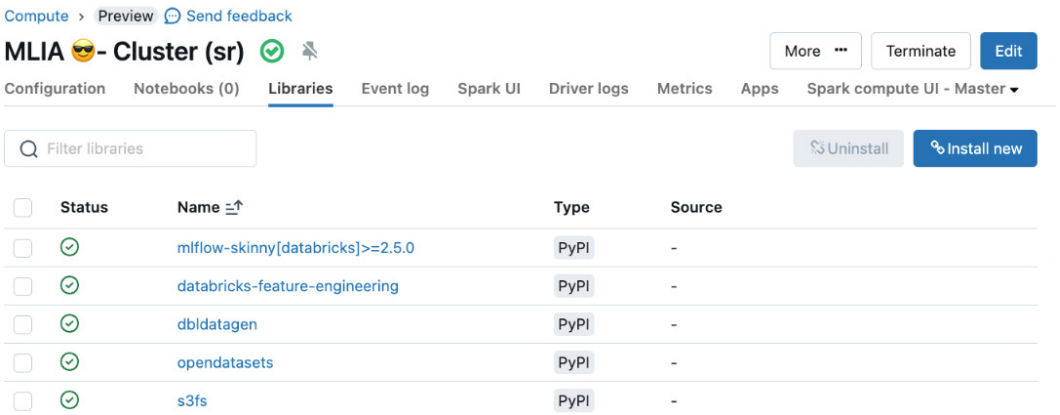


Figure 2.13 – Adding libraries to the Machine Learning in Action (MLIA) cluster configuration